RELATIONAL SCHEMA: SMART HOSTEL MANAGEMENT SYSTEM

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Section: C

Task 1:

1. HOSTEL(Hostel_ID PK, Hostel_Name, Location, Capacity)

2. STAFF(Staff_ID PK, Name, Salary, Designation,
   Hostel_ID FK -> HOSTEL(Hostel_ID) NOT NULL,
   Supervisor_ID FK -> STAFF(Staff_ID) NULL)

   [Step 6 - HOSTEL EMPLOYS STAFF (1:N, Staff=total) -> FK on STAFF]
   [Step 8 - STAFF SUPERVISES STAFF (recursive 1:N, partial) -> FK on same table]

3. ROOM(Hostel_ID FK -> HOSTEL(Hostel_ID), Room_Number, Floor,
   PK (Hostel_ID, Room_Number),
   ON DELETE CASCADE)

   [Step 4 - weak entity ROOM identified by owner HOSTEL via identifying relationship HAS -> composite PK = owner PK + partial key]

4. BED(Hostel_ID, Room_Number, Bed_Number, Availability,
   PK (Hostel_ID, Room_Number, Bed_Number),
   FK (Hostel_ID, Room_Number) -> ROOM(Hostel_ID, Room_Number),
   ON DELETE CASCADE)

   [Step 5 - multivalued composite attribute BED of ROOM -> own relation]

```
 5. STUDENT(Student_ID PK, First_Name, Middle_Name, Last_Name,
    Gender, Date_of_Birth, Email_Address)

    [Step 2 - composite attribute Student_Name -> flattened into components]
    [Age is derived -> not stored]

 6. STUDENT_PHONE(Student_ID FK -> STUDENT(Student_ID), Phone_Number,
    PK (Student_ID, Phone_Number))

    [Step 5 - multivalued attribute Phone_Number -> own relation]

 7. ALLOCATES(Hostel_ID, Room_Number, Student_ID, Allocation_Date,
    Check_in_Date, Check_out_Date,
    PK (Hostel_ID, Room_Number, Student_ID, Allocation_Date),
    FK (Hostel_ID, Room_Number) -> ROOM(Hostel_ID, Room_Number),
    FK Student_ID -> STUDENT(Student_ID))

    [Step 7 - ROOM ALLOCATES STUDENT (M:N) -> new relation with FKs of
    both participants + relationship attributes]

 8. PAYMENT(Payment_ID PK, Payment_Date, Amount, Payment_Type,
    Student_ID FK -> STUDENT(Student_ID) NOT NULL)

    [Step 6 - STUDENT MAKES PAYMENT (1:N, Payment=total) -> FK on PAYMENT]

 9. RECEIPT(Receipt_ID PK, Receipt_Date,
    Payment_ID FK -> PAYMENT(Payment_ID) NOT NULL UNIQUE,
    ON DELETE CASCADE)

    [Step 3 - PAYMENT GENERATES RECEIPT (1:1, both total) -> FK placed on
    RECEIPT since Receipt cannot exist without Payment]

10. FACILITY(Facility_ID PK, Facility_Name, Operating_Hours,
    Hostel_ID FK -> HOSTEL(Hostel_ID) NOT NULL)

    [Step 6 - HOSTEL PROVIDES FACILITY (1:N, Facility=total) -> FK on
    FACILITY]

11. USES(Student_ID, Facility_ID, Usage_Date,
    PK (Student_ID, Facility_ID, Usage_Date),
    FK Student_ID -> STUDENT(Student_ID),
    FK Facility_ID -> FACILITY(Facility_ID))
```

```
[Step 7 - STUDENT USES FACILITY (M:N) -> new relation with FKs of
both participants]
```

TASK 2:

0. Create database:

```sql
CREATE DATABASE smart_hostel_mgmt;
USE smart_hostel_mgmt;
```

```
(PES2UG24CS157@root@localhost) [(none)] > CREATE DATABASE smart_hostel_mgmt;
Query OK, 1 row affected (0.04 sec)

(PES2UG24CS157@root@localhost) [(none)] > use smart_hostel_mgmt;
Database changed
(PES2UG24CS157@root@localhost) [smart_hostel_mgmt] >
```

1. Create all tables:

```sql
CREATE TABLE HOSTEL (
    Hostel_ID      INT PRIMARY KEY,
    Hostel_Name    VARCHAR(50) NOT NULL,
    Location       VARCHAR(100),
    Capacity       INT CHECK (Capacity > 0)
);

CREATE TABLE STAFF (
    Staff_ID        INT PRIMARY KEY,
    Name            VARCHAR(50) NOT NULL,
    Salary          DECIMAL(10,2) DEFAULT 0,
    Job             VARCHAR(50),
    Hostel_ID       INT NOT NULL,
    Supervisor_ID   INT,
    FOREIGN KEY (Hostel_ID) REFERENCES HOSTEL(Hostel_ID)
        ON DELETE CASCADE ON UPDATE CASCADE,
    FOREIGN KEY (Supervisor_ID) REFERENCES STAFF(Staff_ID)
        ON DELETE SET NULL ON UPDATE CASCADE
);
...
...
```

```
(PES2UG24CS157@root@localhost) [smart_hostel_mgmt] > CREATE TABLE HOSTEL (    Hostel_ID      INT PRIMARY KEY,    Hostel
_Name    VARCHAR(50) NOT NULL,    Location       VARCHAR(100),    Capacity       INT CHECK (Capacity > 0));
Query OK, 0 rows affected (0.03 sec)

(PES2UG24CS157@root@localhost) [smart_hostel_mgmt] > CREATE TABLE STAFF (    Staff_ID        INT PRIMARY KEY,    Name
        VARCHAR(50) NOT NULL,    Salary          DECIMAL(10,2) DEFAULT 0,    Job             VARCHAR(50),    Hostel_ID
     INT NOT NULL,    Supervisor_ID   INT,    FOREIGN KEY (Hostel_ID) REFERENCES HOSTEL(Hostel_ID)        ON DELETE CAS
CADE ON UPDATE CASCADE,    FOREIGN KEY (Supervisor_ID) REFERENCES STAFF(Staff_ID)        ON DELETE SET NULL ON UPDATE
CASCADE);
Query OK, 0 rows affected (0.03 sec)

(PES2UG24CS157@root@localhost) [smart_hostel_mgmt] > CREATE TABLE ROOM (    Hostel_ID    INT,    Room_Number  VARCHAR(
10),    Floor        INT,    PRIMARY KEY (Hostel_ID, Room_Number),    FOREIGN KEY (Hostel_ID) REFERENCES HOSTEL(Hostel
_ID)        ON DELETE CASCADE ON UPDATE CASCADE);
Query OK, 0 rows affected (0.02 sec)

(PES2UG24CS157@root@localhost) [smart_hostel_mgmt] > CREATE TABLE BED (    Hostel_ID    INT,    Room_Number   VARCHAR
(10),    Bed_Number    VARCHAR(10),    Availability  ENUM('Available','Occupied') DEFAULT 'Available',    PRIMARY KEY
(Hostel_ID, Room_Number, Bed_Number),    FOREIGN KEY (Hostel_ID, Room_Number) REFERENCES ROOM(Hostel_ID, Room_Number)
       ON DELETE CASCADE ON UPDATE CASCADE);
Query OK, 0 rows affected (0.03 sec)

(PES2UG24CS157@root@localhost) [smart_hostel_mgmt] > CREATE TABLE STUDENT (    Student_ID     INT PRIMARY KEY,    Firs
t_Name     VARCHAR(50) NOT NULL,    Middle_Name    VARCHAR(50),    Last_Name      VARCHAR(50) NOT NULL,    Gender
   CHAR(1),    Date_of_Birth  DATE,    Email_Address  VARCHAR(100) UNIQUE);
Query OK, 0 rows affected (0.06 sec)
```

```
(PES2UG24CS157@root@localhost) [smart_hostel_mgmt] > show tables;
+----------------------------+
| Tables_in_smart_hostel_mgmt |
+----------------------------+
| allocates                  |
| bed                        |
| facility                   |
| hostel                     |
| payment                    |
| receipt                    |
| room                       |
| staff                      |
| student                    |
| student_phone              |
| uses                       |
+----------------------------+
11 rows in set (0.01 sec)
```

2. Alter FACILITY

```sql
ALTER TABLE FACILITY ADD COLUMN Facility_Code VARCHAR(10) FIRST;
DESC FACILITY;
```

```
(PES2UG24CS157@root@localhost) [smart_hostel_mgmt] > ALTER TABLE FACILITY ADD COLUMN Facility_Code VARCHAR(10) FIRST;
Query OK, 0 rows affected (0.06 sec)
Records: 0  Duplicates: 0  Warnings: 0

(PES2UG24CS157@root@localhost) [smart_hostel_mgmt] > DESC FACILITY;
+-----------------+-------------+------+-----+---------+-------+
| Field           | Type        | Null | Key | Default | Extra |
+-----------------+-------------+------+-----+---------+-------+
| Facility_Code   | varchar(10) | YES  |     | NULL    |       |
| Facility_ID     | int         | NO   | PRI | NULL    |       |
| Facility_Name   | varchar(50) | NO   |     | NULL    |       |
| Operating_Hours | varchar(30) | YES  |     | NULL    |       |
| Hostel_ID       | int         | NO   | MUL | NULL    |       |
+-----------------+-------------+------+-----+---------+-------+
5 rows in set (0.01 sec)
```

3. Add constraint

```sql
ALTER TABLE STAFF ADD CONSTRAINT chk_salary CHECK (Salary >= 0);
SHOW CREATE TABLE STAFF;
```

```
| STAFF | CREATE TABLE `staff` (
  `Staff_ID` int NOT NULL,
  `Name` varchar(50) NOT NULL,
  `Salary` decimal(10,2) DEFAULT '0.00',
  `Job` varchar(50) DEFAULT NULL,
  `Hostel_ID` int NOT NULL,
  `Supervisor_ID` int DEFAULT NULL,
  PRIMARY KEY (`Staff_ID`),
  KEY `Hostel_ID` (`Hostel_ID`),
  KEY `Supervisor_ID` (`Supervisor_ID`),
  CONSTRAINT `staff_ibfk_1` FOREIGN KEY (`Hostel_ID`) REFERENCES `hostel` (`Hostel_ID`) ON DELETE CASCADE ON UPD
SCADE,
  CONSTRAINT `staff_ibfk_2` FOREIGN KEY (`Supervisor_ID`) REFERENCES `staff` (`Staff_ID`) ON DELETE SET NULL ON
 CASCADE,
  CONSTRAINT `chk_salary` CHECK ((`Salary` >= 0))
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
+-------+------------------------------------------------------------------------------------------------
--------------------------------------------------------------------------------------------------------
--------------------------------------------------------------------------------------------------------
--------------------------------------------------------------------------------------------------------
```

4. Rename column: Job → Designation (2.4)

```sql
ALTER TABLE STAFF CHANGE COLUMN Job Designation VARCHAR(50);
DESC STAFF;
```

```
(PES2UG24CS157@root@localhost) [smart_hostel_mgmt] > DESC STAFF;
+---------------+---------------+------+-----+---------+-------+
| Field         | Type          | Null | Key | Default | Extra |
+---------------+---------------+------+-----+---------+-------+
| Staff_ID      | int           | NO   | PRI | NULL    |       |
| Name          | varchar(50)   | NO   |     | NULL    |       |
| Salary        | decimal(10,2) | YES  |     | 0.00    |       |
| Designation   | varchar(50)   | YES  |     | NULL    |       |
| Hostel_ID     | int           | NO   | MUL | NULL    |       |
| Supervisor_ID | int           | YES  | MUL | NULL    |       |
+---------------+---------------+------+-----+---------+-------+
6 rows in set (0.00 sec)
```

5. Change Gender to ENUM (2.5)

```sql
ALTER TABLE STUDENT MODIFY COLUMN Gender ENUM('M','F','Other');
DESC STUDENT;
```

```
(PES2UG24CS157@root@localhost) [smart_hostel_mgmt] > DESC STUDENT;
+---------------+------------------------+------+-----+---------+-------+
| Field         | Type                   | Null | Key | Default | Extra |
+---------------+------------------------+------+-----+---------+-------+
| Student_ID    | int                    | NO   | PRI | NULL    |       |
| First_Name    | varchar(50)            | NO   |     | NULL    |       |
| Middle_Name   | varchar(50)            | YES  |     | NULL    |       |
| Last_Name     | varchar(50)            | NO   |     | NULL    |       |
| Gender        | enum('M','F','Other')  | YES  |     | NULL    |       |
| Date_of_Birth | date                   | YES  |     | NULL    |       |
| Email_Address | varchar(100)           | YES  | UNI | NULL    |       |
+---------------+------------------------+------+-----+---------+-------+
7 rows in set (0.00 sec)
```

6. Drop the renamed column (2.6)

```sql
ALTER TABLE STAFF DROP COLUMN Designation;
DESC STAFF;
```

```
(PES2UG24CS157@root@localhost) [smart_hostel_mgmt] > DESC STAFF;
+---------------+---------------+------+-----+---------+-------+
| Field         | Type          | Null | Key | Default | Extra |
+---------------+---------------+------+-----+---------+-------+
| Staff_ID      | int           | NO   | PRI | NULL    |       |
| Name          | varchar(50)   | NO   |     | NULL    |       |
| Salary        | decimal(10,2) | YES  |     | 0.00    |       |
| Hostel_ID     | int           | NO   | MUL | NULL    |       |
| Supervisor_ID | int           | YES  | MUL | NULL    |       |
+---------------+---------------+------+-----+---------+-------+
5 rows in set (0.00 sec)
```

7. Change a column’s data size (2.7)

```sql
ALTER TABLE HOSTEL MODIFY COLUMN Hostel_Name VARCHAR(100);
DESC HOSTEL;
```

```
(PES2UG24CS157@root@localhost) [smart_hostel_mgmt] > ALTER TABLE HOSTEL MODIFY COLUMN Hostel_Name VARCHAR(100);
Query OK, 0 rows affected (0.06 sec)
Records: 0  Duplicates: 0  Warnings: 0

(PES2UG24CS157@root@localhost) [smart_hostel_mgmt] > DESC HOSTEL;
+-------------+--------------+------+-----+---------+-------+
| Field       | Type         | Null | Key | Default | Extra |
+-------------+--------------+------+-----+---------+-------+
| Hostel_ID   | int          | NO   | PRI | NULL    |       |
| Hostel_Name | varchar(100) | YES  |     | NULL    |       |
| Location    | varchar(100) | YES  |     | NULL    |       |
| Capacity    | int          | YES  |     | NULL    |       |
+-------------+--------------+------+-----+---------+-------+
4 rows in set (0.00 sec)
```

8. Rename a table (2.8)

```sql
ALTER TABLE PAYMENT RENAME TO FEE_PAYMENT;
SHOW TABLES;
```

```
(PES2UG24CS157@root@localhost) [smart_hostel_mgmt] > ALTER TABLE PAYMENT RENAME TO FEE_PAYMENT;
Query OK, 0 rows affected (0.03 sec)

(PES2UG24CS157@root@localhost) [smart_hostel_mgmt] > SHOW TABLES;
+-----------------------------+
| Tables_in_smart_hostel_mgmt |
+-----------------------------+
| allocates                   |
| bed                         |
| facility                    |
| fee_payment                 |
| hostel                      |
| receipt                     |
| room                        |
| staff                       |
| student                     |
| student_phone               |
| uses                        |
+-----------------------------+
11 rows in set (0.00 sec)
```

9. Drop tables after dropping constraints (2.9)

```sql
SHOW CREATE TABLE RECEIPT;
```

```
+---------+------------------------------------------------------------------------------------------------------
------------------------------------------------------------------------------------------------------------------
------------------------------------------------------------------------------------------------------------------
-------------------------------------------------+
| RECEIPT | CREATE TABLE `receipt` (
  `Receipt_ID` int NOT NULL,
  `Receipt_Date` date DEFAULT NULL,
  `Payment_ID` int NOT NULL,
  PRIMARY KEY (`Receipt_ID`),
  UNIQUE KEY `Payment_ID` (`Payment_ID`),
  CONSTRAINT `receipt_ibfk_1` FOREIGN KEY (`Payment_ID`) REFERENCES `fee_payment` (`Payment_ID`) ON DELETE CASCADE ON
UPDATE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
+---------+------------------------------------------------------------------------------------------------------
------------------------------------------------------------------------------------------------------------------
------------------------------------------------------------------------------------------------------------------
-------------------------------------------------+
1 row in set (0.00 sec)

(PES2UG24CS157@root@localhost) [smart_hostel_mgmt] >
```

```sql
ALTER TABLE RECEIPT DROP FOREIGN KEY receipt_ibfk_1;
DROP TABLE FEE_PAYMENT;
SHOW TABLES;
```

```
(PES2UG24CS157@root@localhost) [smart_hostel_mgmt] > SHOW TABLES;
+-----------------------------+
| Tables_in_smart_hostel_mgmt |
+-----------------------------+
| allocates                   |
| bed                         |
| facility                    |
| hostel                      |
| receipt                     |
| room                        |
| staff                       |
| student                     |
| student_phone               |
| uses                        |
+-----------------------------+
10 rows in set (0.00 sec)
```